

Build Source

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OpenDingux for Miyoo/Bittboy/PocketGo Handheld

Introduction

- This guide was made by **Steward** and describes how to build all of source codes that include uboot, kernel and emulators

Prepare environment

- Debian 9 (x64)

🔗 Configure toolchain

- Extract toolchain.7z into /opt/miyoo
 - Download form [here](#)
 - NOTE: Is recommended to build your own toolchain due it has received many updates, see [here](#)
- Export command
 - export PATH=\$PATH:/opt/miyoo/bin

Build UBoot

- Boot from SPI Flash
 - make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-licheepi_nano_spiflash_defconfig && make ARCH=arm
- Boot from SDCard
 - make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-licheepi_nano_defconfig && make ARCH=arm

Build Kernel (4bit sdcard - Hardware rev2 and later)

- vim arch/arm/boot/dts/suniv-f1c500s-miyoo.dts +55
 - bus-width = <4>;
- make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- miyoo_defconfig && make ARCH=arm zImage

Build Kernel (1bit sdcard - Hardware rev1)

- vim arch/arm/boot/dts/suniv-f1c500s-miyoo.dts +55
 - bus-width = <1>;
- make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- miyoo_defconfig && make ARCH=arm zImage

Build Boot.scr

- Create a file named boot.cmd and copy this:

```
setenv bootargs console=tty0 console=ttyS1,115200 panic=5 rootwait
root=/dev/mmcblk0p2 rw
load mmc 0:1 0x80C00000 suniv-f1c500s-miyoo.dtb
load mmc 0:1 0x80008000 zImage
bootz 0x80008000 - 0x80C00000
```

- Build the file with:
 - mkimage -C none -A arm -T script -d boot.cmd boot.scr

Prepare SDCard (> = 4GB)

- Partition 1: 256MB FAT32 (boot.scr, dtb and zImage)
- Partition 2: 256MB EXT4 (rootfs)
- Partition 3: 256MB SWAP
- Partition 4: FAT32 (GMenu2X, config files and emulators)

Flash UBoot:

- Boot from SPI Flash (Hardware rev1 and rev2)
 - Short SPI Pin1 and Pin2
 - Connect USB to PC
 - Found device:
 - usb 4-1.2.4.4: New USB device found, idVendor=1f3a, idProduct=efe8

- Release SPI Pin1 and Pin2
- Flash command:
 - `$ sudo sunxi-fel -p spiflash-write 0 u-boot-sunxi-with-spl.bin`
- Boot from SDCard
 - `$ sudo dd if=u-boot-sunxi-with-spl.bin of=/dev/sdX bs=1024 seek=8`

Flash Kernel

- Copy boot.scr into Partition 1
- Copy zImage into Partition 1
- Copy suniv-f1c500s-miyoo.dtb into Partition 1
- Copy r61520fb.ko into kernel folder in Partition 2
- Copy daemon into kernel folder in Partition 2

Build Rootfs

- You can use an already configured buildroot from <https://github.com/MiyooCFW/buildroot>
 - See here: <https://github.com/TriForceX/MiyooCFW/wiki/Making-Games#step-2>
- Or download buildroot-2018.02.9 from <https://buildroot.org>
- Use config_buildroot-2018.02.9 and then make it
- Toolchain location: /opt/miyoo
- Rootfs location: output/images/rootfs.tar

Flash Rootfs

- Extract rootfs.tar into Partition 2

Author website

- You may find other resources and stuff for this (and the other guides) in <https://github.com/steward-fu/miyoo/releases>
- More reviews and details in <https://steward-fu.github.io/website/handheld.htm>

This wiki was made for **stock & custom firmware** usage and tools. If we missed something or if you have a question/suggestion, please send us a message in our *Discord* server [here](#)

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<https://github.com/TriForceX/MiyooCFW.wiki.git>

